In [1]:
```
import pandas as pd
import numpy as np
from sklearn import preprocessing
```

In [5]:
```
students={
    "name":["john","joy","mathew","ron","henry"],
    "Maths":[88,99,np.nan,86,78],
    "Reading":[67,90,130,89,30],
    "Writing":[55,180,np.nan,43,91],
    "Placement":[86,90,45,np.nan,20],
    "Gender":["Male",np.nan,"Male","Male","Female"],
    "JoiningDate":[2018,2022,2024,2005,2013]
}
```

In [6]:
```
df=pd.DataFrame(students)
```

In [7]:
```
df
```

Out[7]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate
0	john	88.0	67	55.0	86.0	Male	2018
1	joy	99.0	90	180.0	90.0	NaN	2022
2	mathew	NaN	130	NaN	45.0	Male	2024
3	ron	86.0	89	43.0	NaN	Male	2005
4	henry	78.0	30	91.0	20.0	Female	2013

In [8]:
```
df.isnull()
```

Out[8]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate
0	False	False	False	False	False	False	False
1	False	False	False	False	False	True	False
2	False	True	False	True	False	False	False
3	False	False	False	False	True	False	False
4	False	False	False	False	False	False	False

In [9]:
```
df.notnull()
```

Out[9]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate
0	True	True	True	True	True	True	True
1	True	True	True	True	True	False	True
2	True	False	True	False	True	True	True
3	True	True	True	True	False	True	True
4	True	True	True	True	True	True	True

In [10]:
```
df
```

Out[10]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate
0	john	88.0	67	55.0	86.0	Male	2018
1	joy	99.0	90	180.0	90.0	NaN	2022
2	mathew	NaN	130	NaN	45.0	Male	2024
3	ron	86.0	89	43.0	NaN	Male	2005
4	henry	78.0	30	91.0	20.0	Female	2013

In [11]:
```
df["Maths"]=df["Maths"].fillna(df["Maths"].mean())
```

In [12]:
```
df
```

Out[12]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate
0	john	88.00	67	55.0	86.0	Male	2018
1	joy	99.00	90	180.0	90.0	NaN	2022
2	mathew	87.75	130	NaN	45.0	Male	2024
3	ron	86.00	89	43.0	NaN	Male	2005
4	henry	78.00	30	91.0	20.0	Female	2013

In [13]:
```
df["Writing"]=df["Writing"].fillna(df["Writing"].median())
```

In [14]:
```
df
```

Out[14]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate
0	john	88.00	67	55.0	86.0	Male	2018
1	joy	99.00	90	180.0	90.0	NaN	2022
2	mathew	87.75	130	73.0	45.0	Male	2024
3	ron	86.00	89	43.0	NaN	Male	2005
4	henry	78.00	30	91.0	20.0	Female	2013

In [15]:
```
df["Placement"]=df["Placement"].fillna(df["Placement"].mode())
```

In [16]:
```
df
```

Out[16]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate
0	john	88.00	67	55.0	86.0	Male	2018
1	joy	99.00	90	180.0	90.0	NaN	2022
2	mathew	87.75	130	73.0	45.0	Male	2024
3	ron	86.00	89	43.0	90.0	Male	2005
4	henry	78.00	30	91.0	20.0	Female	2013

In [17]:
```
df["Writing"]=np.where(df["Writing"]>100,df["Writing"].mean(),df["Writing"])
```

In [18]:
```
df
```

Out[18]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate
0	john	88.00	67	55.0	86.0	Male	2018
1	joy	99.00	90	88.4	90.0	NaN	2022
2	mathew	87.75	130	73.0	45.0	Male	2024
3	ron	86.00	89	43.0	90.0	Male	2005
4	henry	78.00	30	91.0	20.0	Female	2013

In [20]:
```
from sklearn.preprocessing import MinMaxScaler
scaler=MinMaxScaler()
df["Maths"]=scaler.fit_transform(df[["Maths"]])
```

In [21]:
```
df
```

Out[21]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate
0	john	0.476190	67	55.0	86.0	Male	2018
1	joy	1.000000	90	88.4	90.0	NaN	2022
2	mathew	0.464286	130	73.0	45.0	Male	2024
3	ron	0.380952	89	43.0	90.0	Male	2005
4	henry	0.000000	30	91.0	20.0	Female	2013

In [22]:

```
df["Reading_log"]=np.log(df["Reading"])
df
```

Out[22]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate	Reading_log
0	john	0.476190	67	55.0	86.0	Male	2018	4.204693
1	joy	1.000000	90	88.4	90.0	NaN	2022	4.499810
2	mathew	0.464286	130	73.0	45.0	Male	2024	4.867534
3	ron	0.380952	89	43.0	90.0	Male	2005	4.488636
4	henry	0.000000	30	91.0	20.0	Female	2013	3.401197

In [23]:

```
df["Writing"]=np.sqrt(df["Writing"])
df
```

Out[23]:

	name	Maths	Reading	Writing	Placement	Gender	JoiningDate	Reading_log
0	john	0.476190	67	7.416198	86.0	Male	2018	4.204693
1	joy	1.000000	90	9.402127	90.0	NaN	2022	4.499810
2	mathew	0.464286	130	8.544004	45.0	Male	2024	4.867534
3	ron	0.380952	89	6.557439	90.0	Male	2005	4.488636
4	henry	0.000000	30	9.539392	20.0	Female	2013	3.401197

In [ ]: